PAPER EXPLAINER

Conformal guarantees for robots that ask

Two papers read for a robot that must decide whether to grasp or to ask first.

2 papers in 2 sections, two slides each. Every result is quoted from its paper, with the page.

Contents →

Two slides per paper: the idea, then how it works

SLIDE 1 · THE IDEA

- Problem: what goes wrong without the paper.
- Idea: the mechanism, in two or three steps.
- Figure: the paper's own, with its page.
- Takeaway: the sentence to remember.

SLIDE 2 · HOW IT WORKS

- Method, with its formula when it has one.
- What it certifies: the event the theorem bounds, the guarantee, its level and assumptions.
- Measures and results; each result carries its page.
- Limits, on the amber rule.
- For us: what the paper means for the reader's work.

TERMS

Conformal prediction: turns a model's scores into a set that holds the right answer with a chosen probability

Calibration set: labelled examples, kept out of training, that fix the threshold

Coverage: the chance that the set holds the true answer

Risk: the expected value of a loss, such as the fraction of a tumour missed

Blue text is a link. The amber rule marks limits. p.N is the page in the paper; the quote behind each result is in the speaker notes.

CONTENTS

How to read ↑

2 papers in 2 sections

01 · ROBOT PLANNERS 04

KnowNo · CoRL 2023 05

02 · FOUNDATIONS 07

Conformal risk control · Preprint 2022 08

01 Robot planners

A language-model planner that asks for help when its choice is uncertain.

1 paper: KnowNo

Robots That Ask For Help: Uncertainty Alignment for Large Language Model Planners

KnowNo · CoRL 2023 · arXiv 2307.01928

PROBLEM

Told to put the bowl in the microwave, with a metal and a plastic bowl on the counter, an LLM planner may confidently pick one, and the metal one is unsafe.

IDEA

- The LLM lists candidate next steps, then scores them as a multiple-choice question.
- Conformal prediction keeps every option that clears a calibrated cut-off. One option left: act. More: ask.

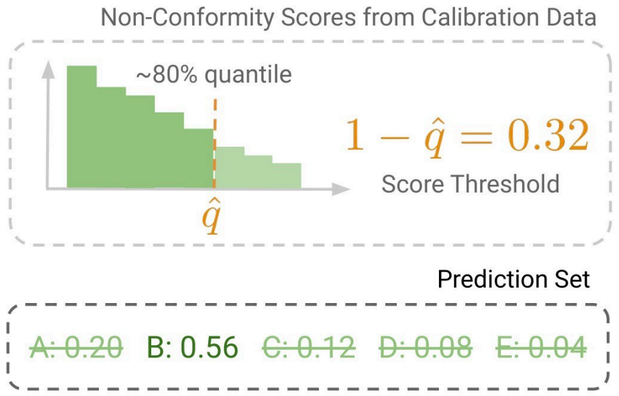


Fig. 2, right panel: calibration sets the score threshold at 0.32; only option B (0.56) clears it: one option left. p.3

> **Score the candidate steps as a multiple-choice question and let conformal prediction set the cut-off: one option left means act, more means ask.**

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The guarantee covers the plan, not the outcome

METHOD

- Candidate steps A to D come from a few-shot prompt; E, "an option not listed here", is added.
- Calibration score: one minus the LLM's probability of the true option, on N = 400 labelled scenarios.

$$C(\tilde{x}) = \left\{ y \in \mathcal{Y} : \hat{f}(\tilde{x})_y \geq 1 - \hat{q} \right\}$$

The set keeps every option the LLM scores at least $1 - \hat{q}$.

MEASURES

- Plan success: the plan is correct, before the robot acts.
- Help rate: the share of trials in which the robot asks.

WHAT IT CERTIFIES

Bounds	the prediction set holds no correct option
Guarantee	coverage, with probability 1 – δ over calibration
Level, n	1 – ε = 0.85 (kitchen); N = 400, δ = 0.01
Assumes	i.i.d. scenarios; the human answers correctly when asked

RESULTS

- Kitchen, PaLM-2L, target 0.85: plan success 0.87, help rate 0.67. p.8
- Simple Set, at equal plan success 0.87: help rate 0.81. p.8

LIMITS

- Execution is outside it: task success is 0.76 at the 0.85 target.

> **For us.** An act-or-ask gate can reuse the one-option rule as it is; to bound wrong grasps, the certified event must be the physical outcome, not the plan.

02 Foundations

The conformal theory behind such a gate, for losses beyond a missed label.

1 paper: Conformal risk control

Contents ↑

Conformal Risk Control

Preprint 2022 · arXiv 2208.02814

PROBLEM

Conformal prediction bounds one error: a set that misses the true label. Many tasks pay a graded loss instead, such as the fraction of a tumour a mask misses.

IDEA

- Index the outputs by λ: a larger λ gives a more conservative output and never a larger loss.
- Choose λ on calibration data so the risk stays at most α even if the next loss is the worst case.

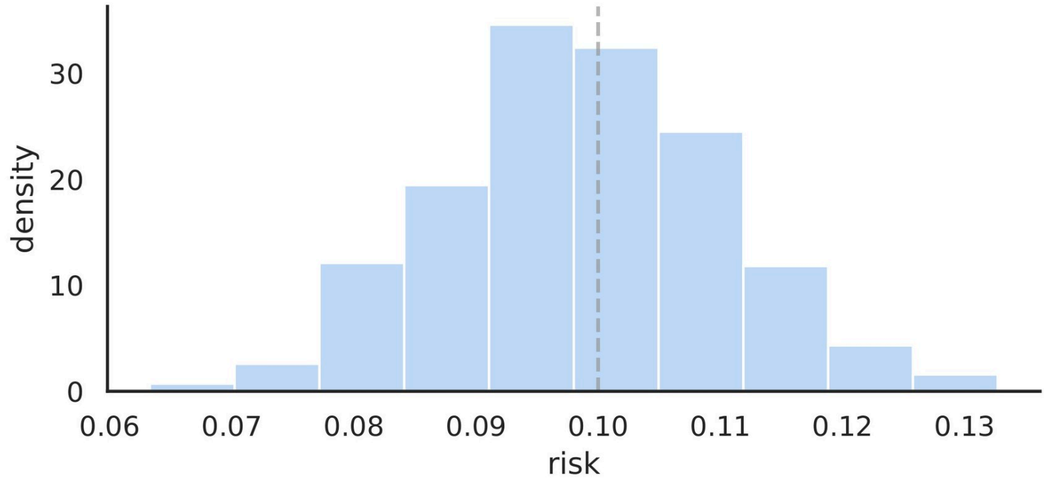


Fig. 1, bottom left: the false-negative rate of the polyp masks over 1000 random splits; α = 0.1 dashed. p. 6

Take the most liberal threshold whose calibration risk, padded by one worst-case loss, stays at most α: the expected loss on a new point is then at most α.

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Expected-loss bound for any bounded monotone loss

METHOD

- Loss: any bounded function of the output that never grows as λ grows, e.g. the fraction of a tumour missed.
- With a 0/1 miss-the-label loss it is split conformal prediction (§2.3).

$$\hat{\lambda} = \inf\left\{\lambda : \frac{n}{n+1}\hat{R}_n(\lambda) + \frac{B}{n+1} \leq \alpha\right\}$$

The smallest λ whose padded calibration loss is at most α.

MEASURES

- Risk: the mean loss on held-out points, per random split.
- Set size: what the guarantee costs in pixels, labels or answers.

WHAT IT CERTIFIES

Bounds	the expected loss on a new test point
Guarantee	in expectation, not with high probability
Level, n	α = 0.1, n = 1000 (polyps); losses at most B = 1
Assumes	exchangeable data; loss monotone and right-continuous in λ

RESULTS

- Polyps, false-negative rate at α = 0.1: mean 0.0987 over 1000 splits. p.6
- ImageNet, graph-distance loss at α = 0.05: mean 0.0499. p.8

LIMITS

- A loss that is not monotone in λ voids it (Prop. 1).

> **For us.** With a 0/1 wrong-grasp loss this rule is split conformal prediction; it adds value once errors are graded, an irreversible one costing more.